Chapter 1

Exercise 1.2.A.

Proof. The two notions can be identified. Indeed, given a group G in the usual sense, define a category with a single object X, and for each $g \in G$, define a morphism $f_g : X \to X$. Further, for two morphisms f_g, f_h, define composition by $f_g \circ f_h = f_{gh}$. This gives a 1-object groupoid, since each morphism is an isomorphism: the morphism f_g has inverse $f_{g^{-1}}$.

Conversely, given a 1-object groupoid, the set of morphisms on that one object forms a group.

On the other hand, there are clearly groupoids that aren't groups, e.g. a category with two objects and only the identity morphisms on those objects. □

Exercise 1.2.B.

Proof. The identity morphism is invertible, with itself as left- and right-inverse. The composition of two invertible morphisms is again invertible, since $(f \circ g) \circ (g^{-1} \circ f^{-1}) = \mathrm{id}$. Finally, the inverse of an invertible morphism is invertible. So, the set of invertible morphisms forms a group under composition.

Given a set X, the automorphisms of X in the category of sets is the set of bijections $f : X \to X$, i.e. the automorphism group is S_X, the symmetric group on X. The automorphisms of a vector space in Vec_k are the automorphisms in the usual sense in linear algebra.

Finally, if A and B are isomorphic objects, with an isomorphism $f : A \to B$, then for an automorphism $\phi : A \to A$, the morphism $f \circ \phi \circ f^{-1}$ is an automorphism of B, with inverse $f \circ \phi^{-1} \circ f^{-1}$. So, this association gives an isomorphism of automorphism groups. □

Exercise 1.2.C. Skipped upon recommendation
Exercise 1.2.D. Skipped upon recommendation
Exercise 1.3.A.

Proof. Let A, B both be initial objects in some category. Then, since A is initial, there is a morphism $f : A \to B$, and since B is initial, there is a morphism $g : B \to A$. Then id_A and $g \circ f$ are both morphisms $A \to A$, but since A is initial, there is a unique such morphism, so $g \circ f = \mathrm{id}_A$. Similarly, $f \circ g = \mathrm{id}_B$. So, f and g are isomorphisms. Further, this isomorphism is unique since any isomorphism $A \to B$ is firstly a morphism, and there is a unique such morphism since A is initial.

The case of final objects is identical. In fact, this proof already shows it by noting that an object is final in a category if and only if it is initial in the opposite category. □

Exercise 1.3.B.

Solution. The initial object in $Sets$ is the empty set, since there is a unique map from the empty set to any other set. The final object in $Sets$ is any set with a single element, since there is a unique map from any set to this set. Note that specifying which single-element set is irrelevant, since any two such sets are uniquely isomorphic as sets.

The initial object in $Rings$ is $\mathbb{Z}$ (recall we are only considering commutative rings with unity and morphisms that preserve unities), since there is a unique morphism given by mapping 1 to $1 \in R$ for a given ring R. The final object is the zero ring.

Top is the "same" as set: the initial object is the empty space and the final object is a single-point space, since there is a unique continuous map in each case.

Given a set S, the poset category of subsets of S has $\emptyset$ as an initial object since it is a subset of each other subset of S, and it has S as a final object. If S is a topological space and we consider the poset category of open subsets of S, the initial and final objects are the same. $\square$

Exercise 1.3.C.

Proof. In one direction, suppose $s \in S$ is a zerodivisor, so that there is some nonzero $x \in A$ with $sx = 0$. Then, since $s(1 \cdot x - 1 \cdot 0)$, we get that $x/1 = 0/1$ in $S^{-1}A$. So, x is in the kernel of the map $A \to S^{-1}A$ and it isn't injective.

Conversely, suppose this map isn't injective. Then, there is some nonzero element x in the kernel, so that $x/1 = 0/1$. I.e. there is some $s \in S$ with $s(1 \cdot x - 1 \cdot 0) = 0$, i.e. $sx = 0$, whence s is a zerodivisor. $\square$

Exercise 1.3.D.

Proof. First, note that $A \to S^{-1}A$ has the property that all elements of S are mapped to invertible elements, since for $s \in S$, the image is $s/1$, which has inverse $1/s \in S^{-1}A$. Now we want to see that it is initial with respect to this property. So, let $f : A \to B$ be a morphism with each element of S mapping to an invertible element of B. Then, we'd like to define a factorization through $S^{-1}A$. Define:

$$g(a/s) = f(a)f(s)^{-1}$$

First, note that this definition makes sense, since for each $s \in S$, $f(s)$ is invertible and so $f(s)^{-1}$ exists. Second, we want to see that it is well-defined. So, suppose $a/s = b/t$, so that there ihs some $r \in S$ with $r(ta - sb) = 0$. Then,

$$0 = f(r(ta - sb)) = f(r)(f(t)f(a) - f(s)f(b))$$

Since $f(r)$ is invertible, we can cancel it, and then multiplying by the inverses of $f(t)$ and $f(s)$ gives:

$$f(a)f(s)^{-1} = f(b)f(t)^{-1}$$

so that g is well-defined. Finally, we want to see it is a ring homomorphism. First, we have $g(1/1) = f(1)f(1)^{-1} = 1$. Second, if $a/s, b/t \in S^{-1}A$, then

$$g(a/s + b/t) = g((at + bs)/(st)) = f(at + bs)f(st)^{-1} = f(a)f(s)^{-1} + f(b)f(t)^{-1} = g(a/s) + g(b/t)$$

and

$$g((a/s)(b/t)) = g((ab)/(st)) = f(ab)f(st)^{-1} = f(a)f(b)f(s)^{-1}f(t)^{-1} = g(a/s)g(b/t)$$

as desired. Thus g is a ring homomorphism. Penultimately, we show it factors f. Let $\iota : A \to S^{-1}A$ be the canonical map. Then, for $a \in A$,

$$g(\iota(a)) = g(a/1) = f(a)f(1)^{-1} = f(a)$$

so that $f = g \circ \iota$ as desired. To finally finish, we need to see that g is unique. So, suppose we have another ring homomorphism $h : S^{-1}A \to B$ with $f = h \circ \iota$. Then, let $a/s \in S^{-1}A$. We have

$$h(a/s) = h((a/1)(1/s)) = h(a/1)h(1/s) = h(a/1)h(s/1)^{-1} = h(\iota(a))h(\iota(s))^{-1} = f(a)f(s)^{-1} = g(a/s)$$

so that $h = g$. This completes the proof that $S^{-1}A$ is initial among such rings.

We address the "furthermore." Let M be an $S^{-1}A$-module. Then, it is also an A-module via $a \cdot m = (a/1) \cdot m$ for $a \in A$ and $m \in M$ (restriction of scalars). For $s \in S$, multiplication by s is an isomorphism of M since it has inverse given by multiplication by $1/s$.

Conversely, suppose M is an A module such that multiplication by s is an automorphism of M for all $s \in S$. For $x \in A$, let $\varphi(x) : M \to M$ denote multiplication by x, and let $R' = \varphi(A)$ be the image of A in the endomorphism ring of M. Then R' is a commutative ring with unity, and an application of Zorn's Lemma gives a maximal subring R of the endomorphism ring of M containing R'. Then, $\varphi : A \to R$ and each element of S maps to an invertible element of R, so this factors as a map from $S^{-1}A$, i.e. we get a map from $S^{-1}A$ into the endomorphism ring of M, i.e. M is an $S^{-1}A$-module. □

Note: This last part of the proof could be simplified if we prove that the universal property doesn't require the target ring to be commutative, because then we can directly note that an A-module M with image of each $s \in S$ being an isomorphism is precisely a map from A into the endomorphism ring of M with each element of S mapping to a unit.

Exercise 1.3.E.

Proof. As suggested, define $S^{-1}M$ by symbols m/s with $m \in M$ and $s \in S$ such that $m/s = n/t$ iff there is some $u \in S$ with $u(tm - sn) = 0$. This is an equivalence relation on pairs (m, s) since $1(sm - sm) = 0$, $u(tm - sn) = 0$ implies $u(sn - tm) = 0$, and if $u(tm - sn) = 0$ and $v(rn - t\ell) = 0$ implies

$$ut(rm - s\ell) = r(utm) - uts\ell = r(usn) - uts\ell = us(rn - t\ell) = 0$$

so that $m/s = \ell/r$ (note that $ut \in S$ since $u, t \in S$ and S is multiplicative).

We have the operation $m/s + n/t = (tm + sn)/(st)$. This is well-defined, for if $m/s = m'/s'$ and $n/t = n'/t'$, so that $u(t'n - tn') = 0 = v(s'm - sm')$, then

$$uv(s't'(tm + sn) - st(t'm' + s'n')) = utt'(v(s'm - sm')) + vss'(u(t'n - tn')) = 0$$

so the sums are equal. Under this operation, $S^{-1}M$ is a group, since $0/1 + m/s = (s0 + 1m)/(1s) = m/s$, and $m/s + (-m)/s = (sm + s(-m))/(s^2) = 0/(s^2) = 0/1$, and:

$$\left(\frac{m}{s} + \frac{n}{t}\right) + \frac{\ell}{r} = \frac{tm + sn}{st} + \frac{\ell}{r} = \frac{r(tm + sn) + st\ell}{st(r)} = \frac{trm + s(rn + t\ell)}{s(tr)} = \frac{m}{s} + \frac{rn + t\ell}{tr} = \frac{m}{s} + \left(\frac{n}{t} + \frac{\ell}{r}\right)$$

Now, $S^{-1}M$ is an A-module, under $a \cdot (m/s) = (am)/s$, where the fact that this satisfies the module conditions follows from the fact that M is an A-module [proof omitted, this is already so long].

Finally, we have the map $\phi : M \to S^{-1}M$ via $m \mapsto m/1$. This has the property that for $s \in S$, multiplication by s is invertible, with inverse $m/t \mapsto m/(ts)$. Further, if N is an A-module with each element of S inducing an automorphism of N and $\alpha : M \to N$ is an A-module homomorphism, then we get a map $\alpha' : S^{-1}M \to N$ via $m/s \mapsto s^{-1} \cdot \alpha(m)$, where s^{-1} denotes the inverse of the isomorphism of N induced by s. □

Exercise 1.3.F.

Proof. Parts (a) and (b) are both subsumed under showing that localization commutes with arbitrary direct sums. For this, it's useful to give direct sums a universal property. We claim (and show after this proof) that the direct sum of A-modules $\{M_j\}$ is an A-module P along with maps $P \to M_j$ that is initial with respect to have maps into each M_j.

Now, given this property, define $P = \bigoplus_j M_j$ and $Q = \bigoplus_j S^{-1}M_j$. We wish to show that $Q = S^{-1}P$. To do this, we show that Q has the universal property of being the localization of P. First, we need a map $P \to Q$. To find it, note that for each j we have maps $\ell_j : M_j \to S^{-1}M_j$ via localization and $g_j : S^{-1}M_j \to Q$ via the direct sum. Hence, since P is the direct sum, we have a unique map $f : P \to Q$ such that $f \circ f_j = g_j \circ \ell_j$ for each j, i.e. we have commutative diagrams:

$$\begin{tikzcd} M_j \arrow[r, "\ell_j"] \arrow[d, "f_j"] & S^{-1}M_j \arrow[d, "g_j"] \\ P \arrow[r, "f"] & Q \end{tikzcd}$$

Finally, we show that $f : P \to Q$ is the localization map. To prove this, let N be an $S^{-1}A$-module, and let $\alpha : P \to N$ be a morphism. Then, for each j, we have the composition map $\alpha \circ f_j$, and by the universal property of the localization, we thus get that each of these factors through a map $\alpha_j : S^{-1}M_j \to N$. I.e. the diagram extends

$$\begin{tikzcd} M_j \arrow[r, "\ell_j"] \arrow[d, "f_j"] & S^{-1}M_j \arrow[d, "g_j"] \arrow[ddr, "\alpha_j"] & \\ P \arrow[r, "f"] \arrow[drr, "\alpha"] & Q \arrow[dr, dashed, "\beta"] & \\ & & N \end{tikzcd}$$

By the universal property of Q, the dashed arrow exists and makes the diagram commute. We finally wish to show that $\alpha = \beta \circ f$. Fix a j, and note that

$$(\beta \circ f) \circ f_j = \beta \circ (g_j \circ \ell_j) = \alpha_j \circ \ell_j = \alpha \circ f_j$$

So, the two maps agree when composed by f_j for each j, and by the uniqueness part of the property of direct sums, we get that $\beta \circ f = \alpha$.

Finally, we wish to show β is unique. Suppose $\gamma : Q \to N$ satisfies $\gamma \circ f = \alpha$. Then, for each j we have

$$\gamma \circ g_j \circ \ell_j = \gamma \circ f \circ f_j = \beta \circ f \circ f_j = \beta \circ g_j \circ \ell_j$$

By the universal property of ℓ_j, for each j, we get $\gamma \circ g_j = \beta \circ g_j$. Then, by the universal property of Q, we have that $\gamma = \beta$. This proves uniqueness. In other words, (Q, f) has the property that for any S-invertible map $P \to N$ factors uniquely through f, so that $Q = S^{-1}P$.

For the counterexample, we consider the suggestion, and let $A = \mathbb{Z}$ and for each $n \in \mathbb{N}$ let $M_n = \mathbb{Z}$ as well. Let $S = \mathbb{Z} \setminus \{0\}$, so that $S^{-1}M_n = \mathbb{Q}$ for each n. Thus,

$$(1, 1/2, 1/3, 1/4, \ldots) \in \prod_{n=1}^{\infty} \mathbb{Q} = Q$$

On the other hand, our explicit description of $S^{-1}P$ shows that for each element of $S^{-1}P$, there is an element of S that multiplies it into (the image of) P. But for this element, there is no multiple of it that consists entirely of integers, since the denominators get arbitrarily large. So, this element of Q isn't in $S^{-1}P$ and so they aren't equal. $\square$

Exercise 1.3.G.

Proof. We'd like to define a map $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/10\mathbb{Z} \otimes \mathbb{Z}/12\mathbb{Z}$, for which we only need to map the single element: let it map to $1 \otimes 1$. This is a homomorphism, since

$$2(1 \otimes 1) = 2(1 \otimes 25) = 10(1 \otimes 5) = 10 \otimes 5 = 0 \otimes 5 = 0$$

It is injective since $1 \otimes 1$ is nonzero. Finally, it is surjective. Indeed, consider an arbitrary element:

$$\sum_j a_j \otimes b_j = \sum_j a_j b_j (1 \otimes 1)$$

which is the image of $\sum_j a_j b_j$. □

Exercise 1.3.H.

Proof. To see that it is a functor, note that it gives a way to map objects to objects, but we also need to map morphisms to morphisms. So, let $f : M \to L$ be a morphism of A-modules. Then, we can define $f \otimes N : M \otimes N \to L \otimes N$ via $m \otimes n \mapsto f(m) \otimes n$ and extending linearly. Now, we have $(\mathrm{id}_M \otimes N)(m \otimes n) = \mathrm{id}_M(m) \otimes n = m \otimes n$ for all m, n, so that the identity is mapped to the identity. Second, if $f : M \to L$ and $g : L \to K$, then for all $m \in M$, $n \in N$:

$$\begin{aligned}((g \circ f) \otimes N)(m \otimes n) &= (g \circ f)(m) \otimes n \\ &= g(f(m)) \otimes n \\ &= (g \otimes N)(f(m) \otimes n) \\ &= (g \otimes N)((f \otimes N)(m \otimes n)) \\ &= ((g \otimes N) \circ (f \otimes N))(m \otimes n)\end{aligned}$$

so that $\otimes N$ distributes over compositions. This shows that $\otimes N$ is a functor.

Now, let $M \xrightarrow{f} L \xrightarrow{g} K \to 0$ is exact. We wish to show that $g \otimes N$ surjects and that $\ker(g \otimes N) = \mathrm{im}(f \otimes N)$. For the first, it suffices for the image to contain each simple tensor, but this is immediate: for $k \in K$ there is $\ell \in L$ with $g(\ell) = k$, so that $(g \otimes N)(\ell \otimes n) = k \otimes n$ for any $n \in N$.

For the second, first note that $((g \otimes N) \circ (f \otimes N)) = (g \circ f) \otimes N = 0 \otimes N = 0$ is the zero map. So, we have that the image is contained in the kernel; we need the reverse. For this, we lift to the free modules lying over the tensor products and pick generators there. We lift to a map $\hat{g} : A[L \times N] \to A[K \times N]$ via $(\ell, n) \mapsto (g(\ell), n)$, and note that $(g \otimes N) \circ \pi_L = \pi_K \circ \hat{g}$, where π_x denotes the projection $A[x \times N] \to x \otimes N$. Since π_L is surjective, we have

$$\ker(g \otimes N) = \pi_L(\ker((g \otimes N) \circ \pi_L)) = \pi_L(\ker(\pi_K \circ \hat{g})) = \pi_L(\hat{g}^{-1}(\ker(\pi_K)))$$

Now, $\ker(\pi_K)$ is generated by all of the tensor relations:

$$(k + k', n) - (k, n) - (k', n), (k, n + n') - (k, n) - (k, n'), a(k, n) - (ak, n), a(k, n) - (k, an)$$

So, the preimage under $\hat{g}$ is generated by the preimage of these generators and the kernel of $\hat{g}$ itself. The preimages are precisely the tensor relations of $L \otimes N$ since g is surjective, i.e. all of these relations:

$$(\ell + \ell', n) - (\ell, n) - (\ell', n), (\ell, n + n') - (\ell, n) - (\ell, n'), a(\ell, n) - (a\ell, n), a(\ell, n) - (\ell, an)$$

Further, $\hat{g}$ is a map between free modules, so the kernel is generated by elements of the form $x - x'$ with x, x' among the basis and $\hat{g}(x) = \hat{g}(x')$. I.e. for elements $\ell, \ell' \in L$ and $n \in N$ with $g(\ell) = g(\ell')$, the kernel is generated by $(\ell, n) - (\ell', n)$.

Finally, we map all of these generators back under π_L. The tensor relations all map to zero in $L \otimes N$ essentially by definition. So, $\ker(g \otimes N)$ is generated by $\ell \otimes n - \ell' \otimes n$ for $g(\ell) = g(\ell')$. But then $g(\ell - \ell') = 0$, so $\ell - \ell' \in \ker(g) = \operatorname{im}(f)$ by exactness, so that there is some $m \in M$ with $f(m) = \ell - \ell'$. But then,

$$(f \otimes N)(m \otimes n) = f(m) \otimes n = (\ell - \ell') \otimes n = \ell \otimes n - \ell' \otimes n$$

is in the image of $(f \otimes N)$. $\square$

Exercise 1.3.I.

Proof. First, suppose $(T', t' : M \times N \to T')$ has the same property, i.e. it is initial among A-bilinear maps from $M \times N$. Then, since (T, t) is initial in this way, we get a map $f : T \to T'$ and since (T', t') is initial, we get a map $g : T' \to T$ such that $t' = f \circ t$ and $t = g \circ t'$. Then,

$$\mathrm{id}_T \circ t = t = g \circ t' = g \circ (f \circ t) = (g \circ f) \circ t$$

and so the uniqueness property of factorizations out of (T, t) gives $\mathrm{id}_T = g \circ f$. Doing the same with t' gives $f \circ g = \mathrm{id}_{T'}$. Thus f and g are isomorphisms. Furthermore, they are uniquely determined since there is a unique map f with $t' = f \circ t$ in the first place. $\square$

Exercise 1.3.J.

Proof. Let $f : M \times N \to L$ be an A-bilinear map and let $t : M \times N \to M \otimes N$ be the usual map $(m, n) \mapsto m \otimes n$. Then f lifts to a map on the free module $\widehat{f} : A[M \times N] \to L$ via $\widehat{f}(x) = f(x)$ for all $x \in M \times N$ and extended A-linearly. Now, I claim that all of the tensor relations are in the kernel of $\widehat{f}$. Indeed, for $m, m' \in M$, $n, n' \in N$, and $a \in A$, we have:

$$\begin{aligned}
&\widehat{f}((m + m', n) - (m, n) - (m', n)) \\
&\qquad = f(m + m', n) - f(m, n) - f(m', n) \\
&\qquad = f(m, n) + f(m', n) - f(m, n) - f(m', n) = 0 \\
&\widehat{f}((m, n + n') - (m, n) - (m, n')) \\
&\qquad = f(m, n + n') - f(m, n) - f(m, n') \\
&\qquad = f(m, n) + f(m, n') - f(m, n) - f(m, n') = 0 \\
&\widehat{f}(a(m, n) - (am, n)) = af(m, n) - f(am, n) = af(m, n) - af(m, n) = 0 \\
&\widehat{f}(a(m, n) - (m, an)) = af(m, n) - f(m, an) = af(m, n) - af(m, n) = 0
\end{aligned}$$

using repeatedly that $\widehat{f}$ extends f via A-linearity and that f is A-bilinear. So, $\widehat{f}$ factors through the quotient by these relations, which is precisely the tensor product we've defined. I.e. there is a map $g : M \otimes N \to L$ with $\widehat{f} = g \circ \pi$, where π is the quotient map.

Now, for $(m, n) \in M \times N$, we have:

$$(g \circ t)(m, n) = g(m \otimes n) = g(\pi(m, n)) = \widehat{f}(m, n) = f(m, n)$$

so that $g \circ t = f$ is the desired factorization. Finally, we wish to show g is unique. Suppose $h : M \otimes N \to L$ also satisfies $h \circ t = f$. Then, for $m \in M$ and $n \in N$,

$$h(m \otimes n) = h(t(m, n)) = f(m, n) = g(t(m, n)) = g(m \otimes n)$$

and so g and h agree on simple tensors. Hence, by linearity, they agree on all of $M \otimes N$, so that $h = g$. $\square$

Exercise 1.3.K.

Proof. Note that f makes B an A-module, and so the tensor $B \otimes_A M$ makes sense and is an A-module. Now, for $b \in B$, define:

$$b \cdot \left(\sum_i b_i \otimes m_i \right) = \sum_i (bb_i) \otimes m_i$$

We want to see that this makes $B \otimes_A M$ into a B-module. It is clear that for $x \in B \otimes M$ and for $b, b' \in B$, that $b \cdot (b' \cdot x) = (bb') \cdot x$ from the definition above and the associativity of multiplication in B. The definition is also clearly linear in $B \otimes M$, since b acts on each summand separately. So, indeed the tensor is a B-module.

We'd like to show this is functorial in M. First, note that if $f : M \to N$ is a morphism of A-modules, then we get a map $\mathrm{id}_B \otimes f : B \otimes_A M \to B \otimes_A N$. By checking on simple tensors, we see that $\mathrm{id}_B \otimes \mathrm{id}_M = \mathrm{id}_{B \otimes M}$. Further, if $f : M \to N$ and $g : N \to L$ are two morphisms of A-modules, then $\mathrm{id}_B \otimes (g \circ f) = (\mathrm{id}_B \otimes g) \circ (\mathrm{id}_B \otimes f)$ by again checking on simple tensors. So this indeed gives a functor from the category of A-modules to that of B-modules.

Now, we give $B \otimes_A C$ a ring structure by defining the product of simple tensors by $(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$ and extending linearly. Since the definition is via linear extension, we immediately get that the product distributes over addition. Further, commutativity and associativity of multiplication follow from commutativity and associativity in each factor. Finally, $1 \otimes 1$ is the unity, so indeed we have a ring structure. $\square$

Exercise 1.3..

Proof. $\square$